PROPOSED METHODOLOGY

System design is the blueprint phase of software engineering that bridges the gap between system requirements and actual implementation. It defines how the system is structured in terms of data flow, architecture, user interface layout, module interaction, and backend schema. This section elaborates on the architectural and component-level designs for the **SHIFTY – An AI Powered**, ensuring smooth integration, scalability, and efficiency of the final deployed system.

4.1. Requirement Analysis

Requirement analysis is the process of identifying the needs and expectations of stakeholders. For **SHIFTY**, the requirements fall under two broad categories: functional (what the system should do) and non-functional (how the system should behave).

4.1.1 Functional Requirements

The functional requirements define the core features and operations that must be present in the system. Below is a detailed breakdown:

User Module

- **Registration and Login**:
  Users can sign up using email, phone number, or third-party providers (Google/Apple). Passwords will be hashed using bcrypt.
- **Profile Management**:
  Users can update their contact information, view booking history, and save frequently used addresses.

Booking Module

- **Create New Booking**: